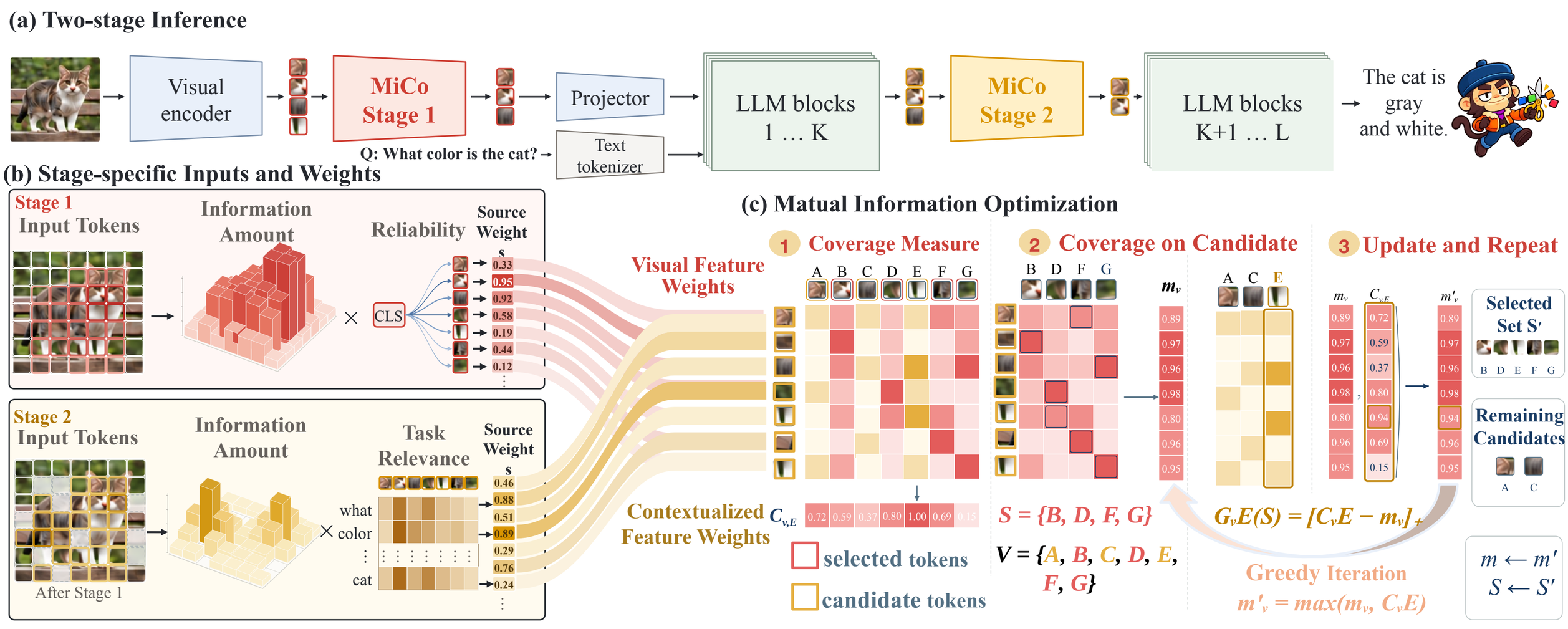

(a) Two-stage Inference
Visual encoder
MiCo Stage 1
Projector
Text tokenizer
Q: What color is the cat?
LLM blocks 1 … K
MiCo Stage 2
LLM blocks K+1 … L
The cat is gray and white.
(b) Stage-specific Inputs and Weights
Stage 1
Input Tokens
Information Amount
Reliability
Source Weights
CLS
0.33
0.95
0.92
0.58
0.19
0.44
0.12
Stage 2
Input Tokens
Information Amount
Task Relevance
Source Weights
what
color
cat
0.46
0.88
0.51
0.89
0.29
0.76
0.24
After Stage 1
(c) Matual Information Optimization
Visual Feature Weights
Contextualized Feature Weights $C_{v,E}$
0.72 0.59 0.37 0.80 1.00 0.69 0.15
selected tokens
candidate tokens
1 Coverage Measure
A B C D E F G
2 Coverage on Candidate
B D F G
m_v
0.89 0.97 0.96 0.98 0.80 0.96 0.95
A C E
3 Update and Repeat
m_v $C_{v,E}$
0.89 0.72
0.97 0.59
0.96 0.37
0.98 0.80
0.80 0.94
0.96 0.69
0.95 0.15
m'_v
0.89 0.97 0.96 0.98 0.94 0.96 0.95
Selected Set S′
B D E F G
Remaining Candidates
A C
$S = \{B, D, F, G\}$
$V = \{A, B, C, D, E, F, G\}$
$G_vE(S) = [C_vE - m_v]_+$
Greedy Iteration
$m'_v = max(m_v, C_vE)$
$m \leftarrow m'$
$S \leftarrow S'$